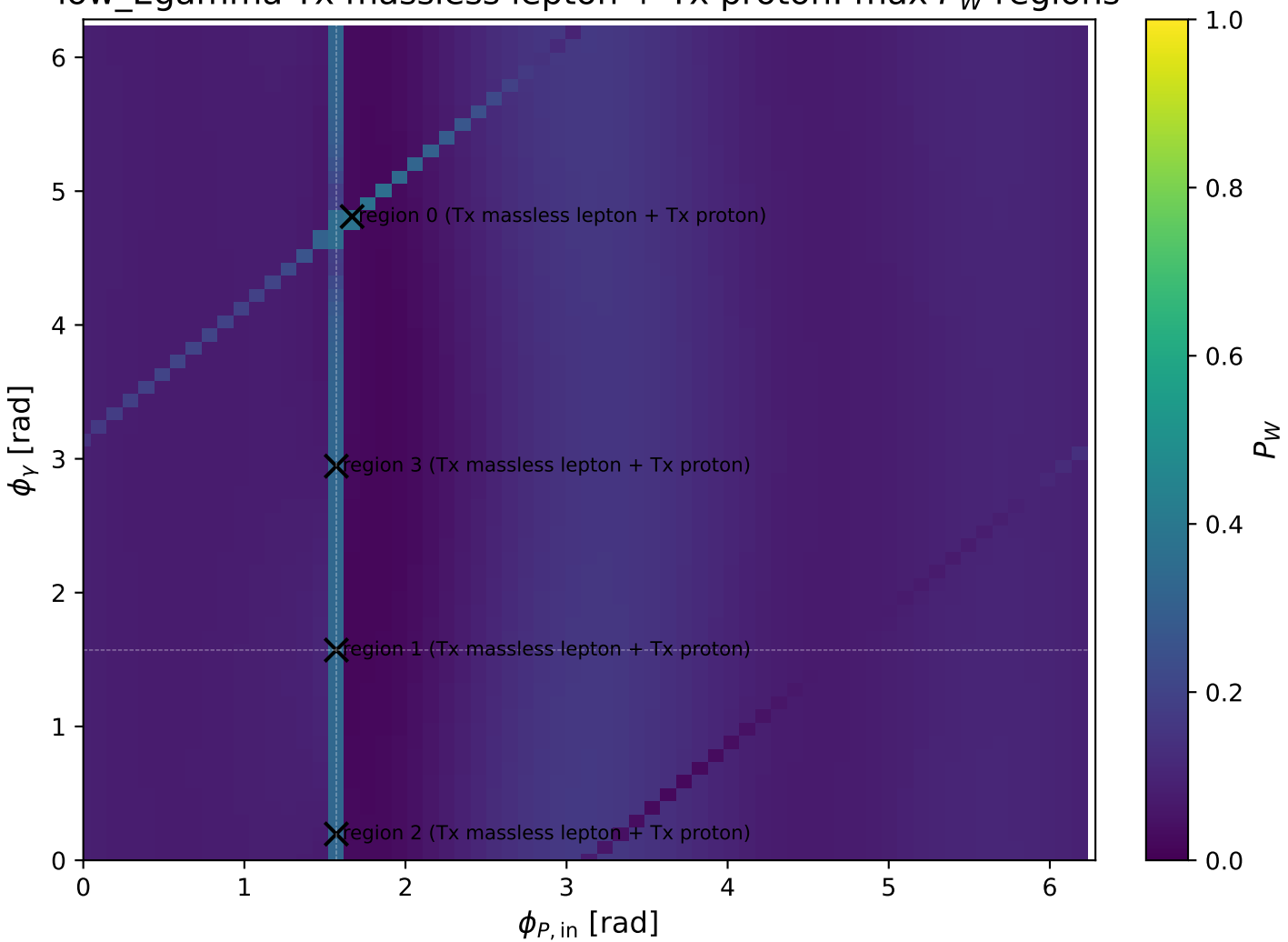
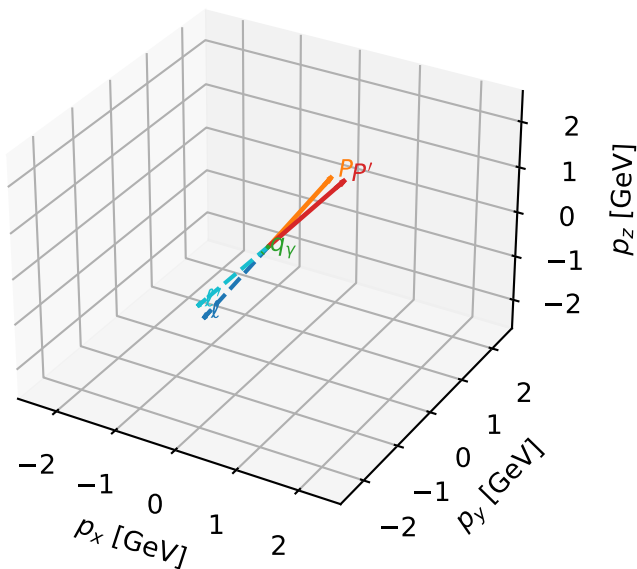


low\_Egamma Tx massless lepton + Tx proton: max  $P_W$  regions



# Tx massless lepton + Tx proton characteristic max $P_W$ configuration

3D momenta

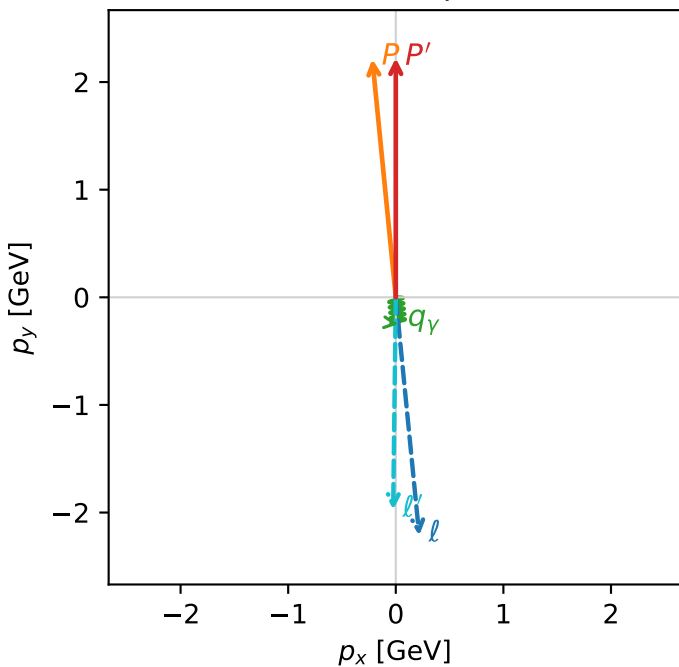


w\_purity\_low\_egamma\_txtx\_region\_0 P\_W=0.382313  
 region: low\_Egamma\_TxTx\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{X,\ell_0}\rangle \otimes |T_{X,p}\rangle$

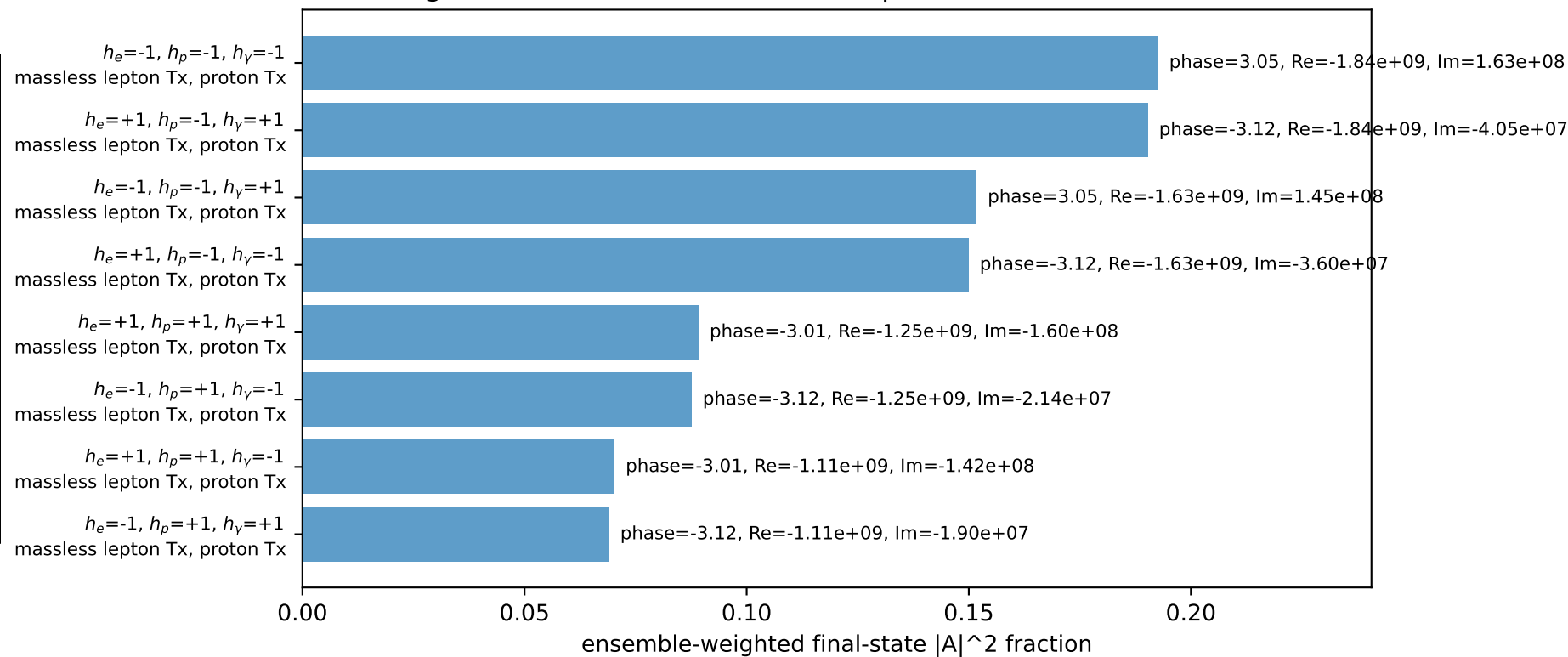
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.22371$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.81056$ ,  $\phi_P=1.66897$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=4.81056$   
 $Q^2=0.053669$ ,  $x_B=0.0226749$ ,  $t=-0.0476372$ ,  $W^2=3.19307$ ,  $y=0.114697$   
 $\theta(\ell', \gamma)=6.33588$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.2180$   $p_y= -2.2137$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.2180$   $p_y= 2.2137$   $p_z= 0.0000$   
 $\ell'$   $E= 1.9751$   $p_x= -0.0245$   $p_y= -1.9749$   $p_z= 0.0000$   
 $P'$   $E= 2.4134$   $p_x= 0.0000$   $p_y= 2.2237$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0245$   $p_y= -0.2488$   $p_z= 0.0000$

Transverse plane

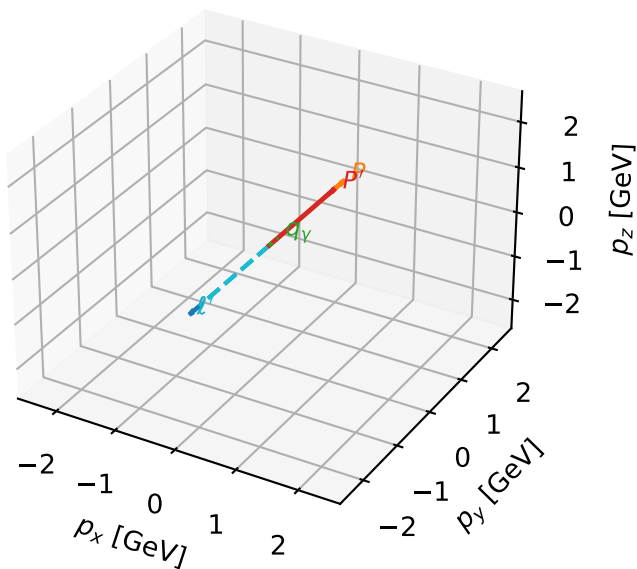


Leading initial-ensemble/final-state components (N=8, retained=100.0%)



# Tx massless lepton + Tx proton characteristic max $P_W$ configuration

3D momenta

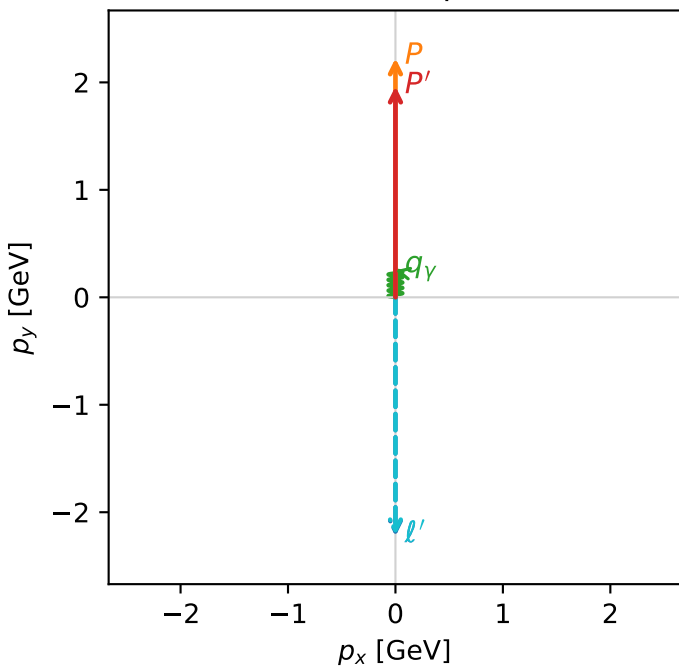


w\_purity\_low\_egamma\_txtx\_region\_1 P\_W=0.333331  
 region: low\_Egamma\_TxTx\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,l_0}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.96296$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=1.5708$   
 $Q^2=8.66557e-12$ ,  $x_B=8.15204e-11$ ,  $t=-0.0114583$ ,  $W^2=0.986143$ ,  $y=0.00515116$   
 $\theta(l', \gamma)=180$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2130$   $p_x= -0.0000$   $p_y= -2.2130$   $p_z= 0.0000$   
 $P'$   $E= 2.1756$   $p_x= 0.0000$   $p_y= 1.9630$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0000$   $p_y= 0.2500$   $p_z= 0.0000$

Transverse plane



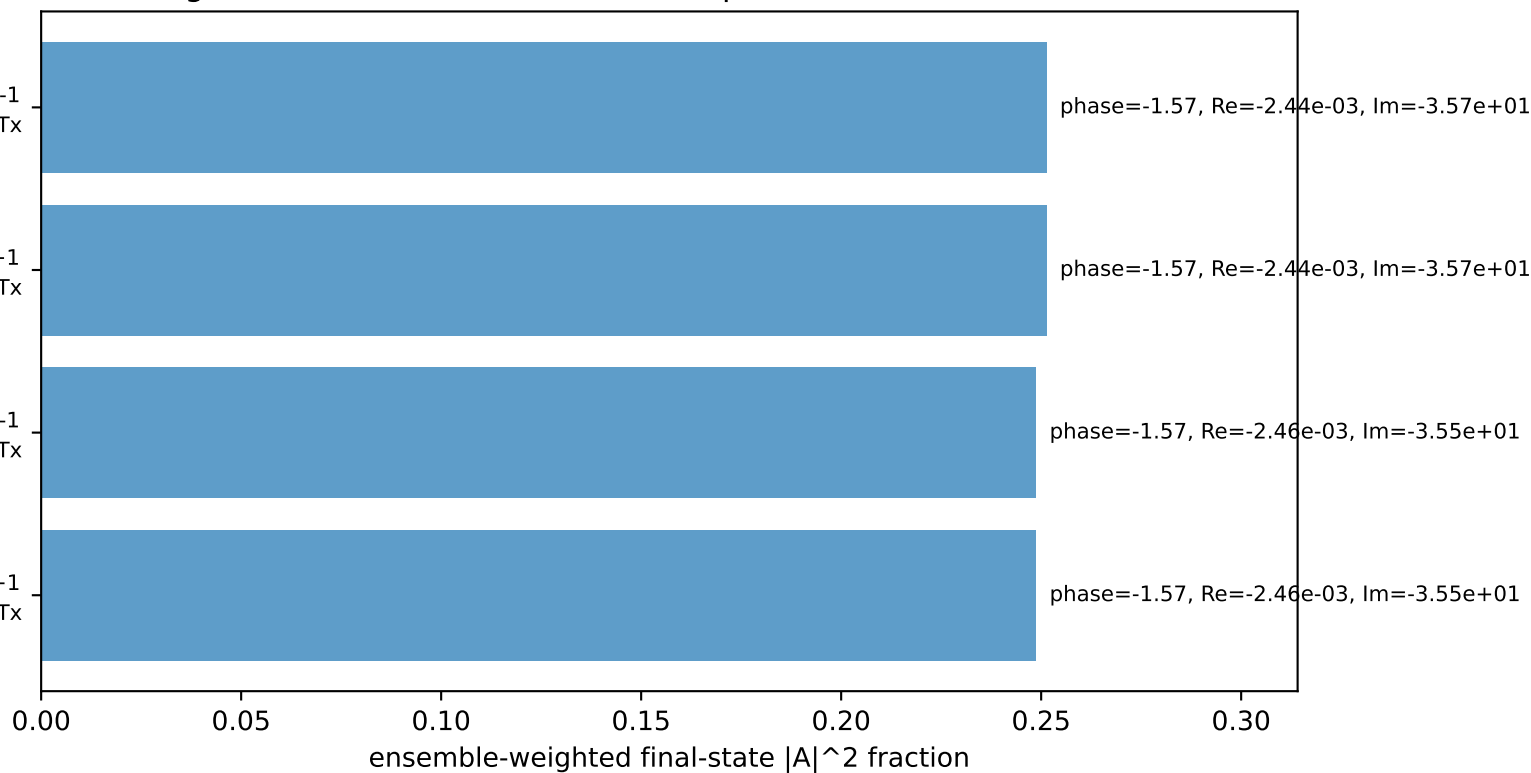
$h_e=-1, h_p=+1, h_\gamma=-1$   
 massless lepton Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=+1$   
 massless lepton Tx, proton Tx

$h_e=+1, h_p=+1, h_\gamma=-1$   
 massless lepton Tx, proton Tx

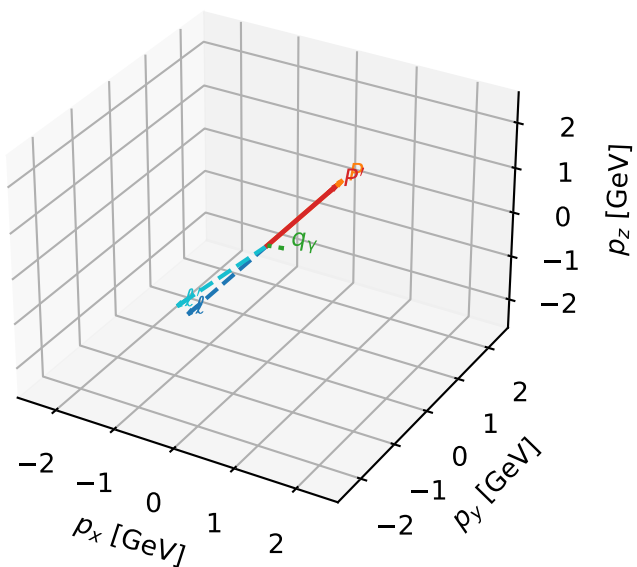
$h_e=-1, h_p=-1, h_\gamma=+1$   
 massless lepton Tx, proton Tx

Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# Tx massless lepton + Tx proton characteristic max $P_W$ configuration

3D momenta

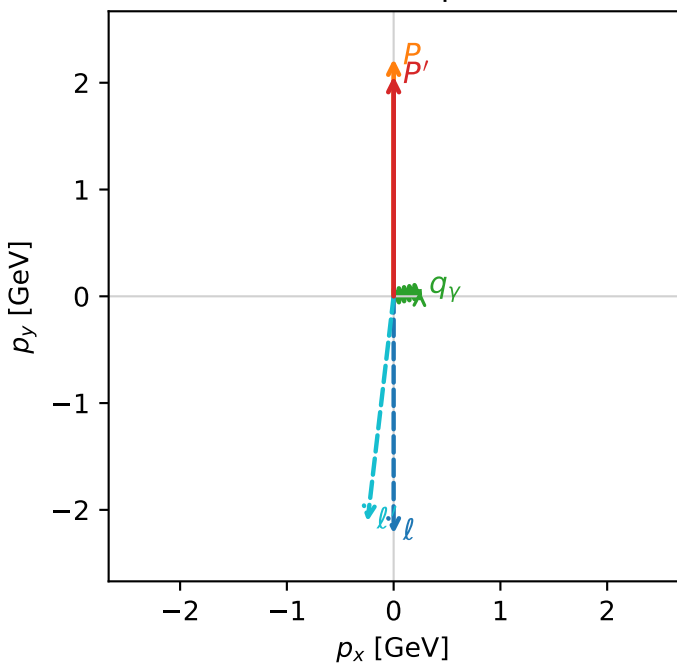


w\_purity\_low\_egamma\_txtx\_region\_2 P\_W=0.328562  
 region: low\_Egamma\_TxTx\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

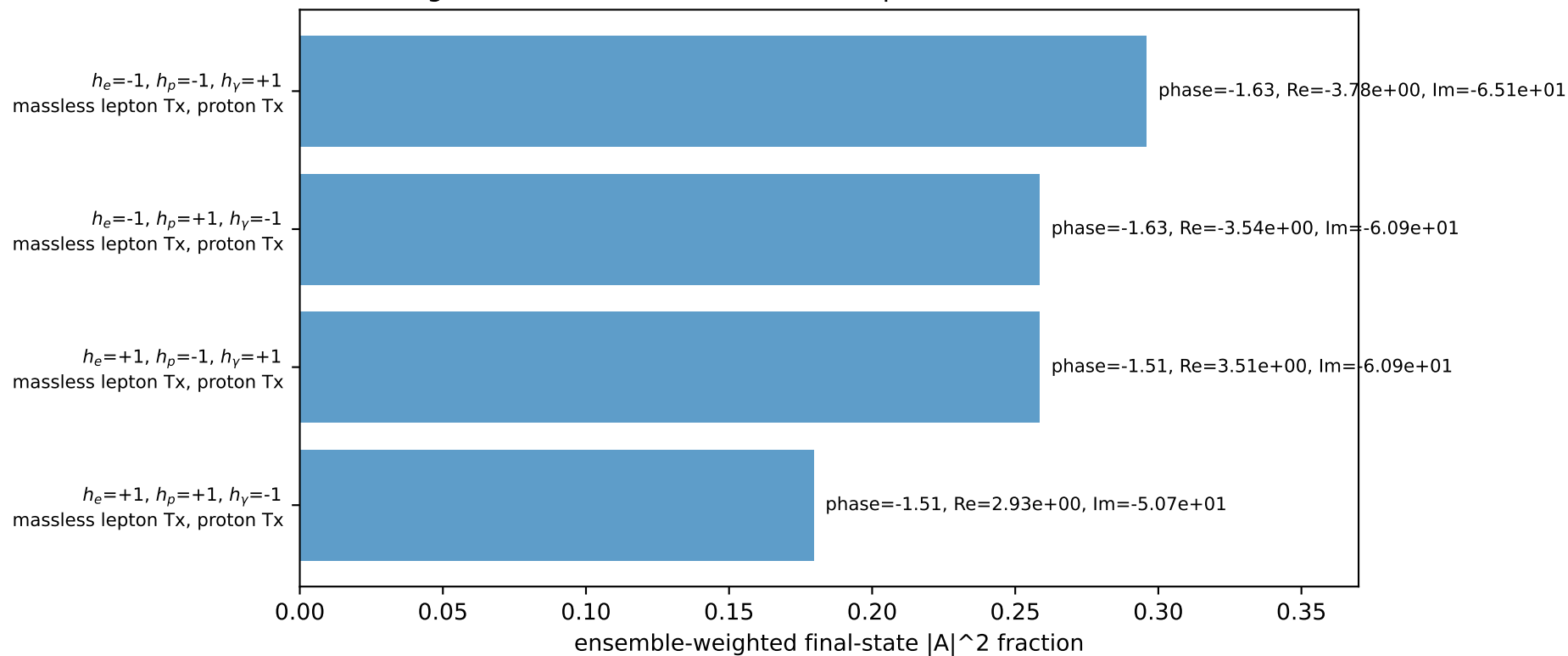
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.06107$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=0.19635$   
 $Q^2=0.0631736$ ,  $x_B=0.0635316$ ,  $t=-0.00429535$ ,  $W^2=1.81103$ ,  $y=0.0481859$   
 $\theta(\ell', \gamma)=107.879$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1240$   $p_x= -0.2452$   $p_y= -2.1098$   $p_z= 0.0000$   
 $P'$   $E= 2.2645$   $p_x= 0.0000$   $p_y= 2.0611$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2452$   $p_y= 0.0488$   $p_z= 0.0000$

Transverse plane

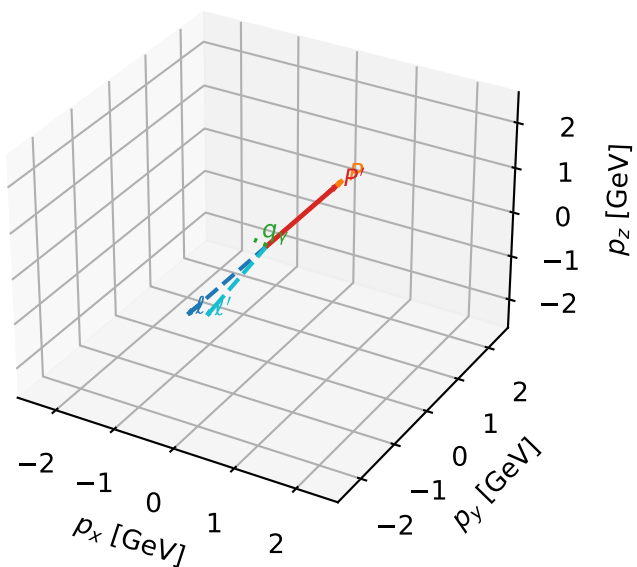


Leading initial-ensemble/final-state components (N=4, retained=99.3%)



# Tx massless lepton + Tx proton characteristic max $P_W$ configuration

3D momenta

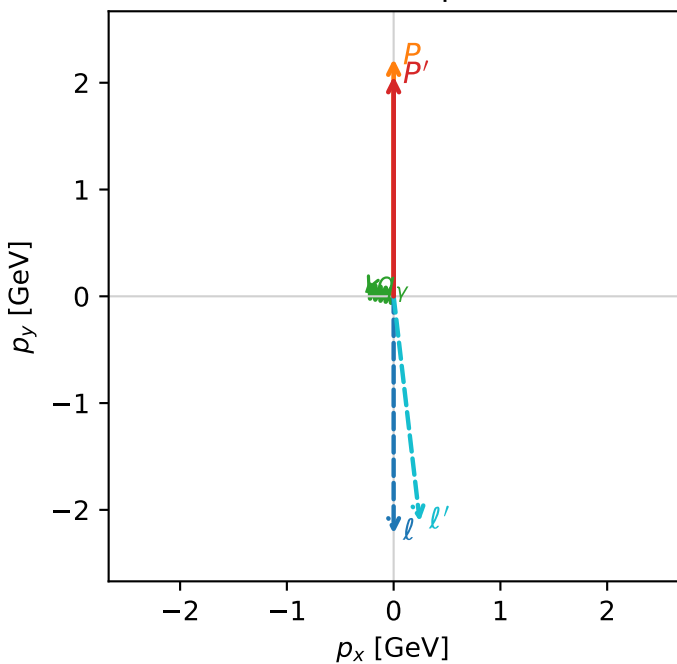


w\_purity\_low\_egamma\_txtx\_region\_3 P\_W=0.328012  
 region: low\_Egamma\_TxTx\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.06107$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=2.94524$   
 $Q^2=0.0631736$ ,  $x_B=0.0635316$ ,  $t=-0.00429535$ ,  $W^2=1.81103$ ,  $y=0.0481859$   
 $\theta(\ell', \gamma)=107.879$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1240$   $p_x= 0.2452$   $p_y= -2.1098$   $p_z= 0.0000$   
 $P'$   $E= 2.2645$   $p_x= 0.0000$   $p_y= 2.0611$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2452$   $p_y= 0.0488$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.3%)

